

05 双极结型三极管及放大电路基础：公式总结

根据 ppt/05 双极结型三极管及放大电路基础.pdf 中出现的主要公式整理。分式、上下标、希腊字母均用数学排版渲染。

1. BJT 放大状态下的电流关系

$$\begin{aligned}I_E &= I_B + I_C \\I_C &= I_{NC} + I_{CBO} \\ \alpha &= \frac{I_{NC}}{I_E}, \quad I_C \simeq \alpha I_E \\ \beta &= \frac{\alpha}{1 - \alpha}, \quad \alpha = \frac{\beta}{1 + \beta} \\ I_C &= \frac{\alpha}{1 - \alpha} I_B + \frac{1}{1 - \alpha} I_{CBO} = \beta I_B + (1 + \beta) I_{CBO} \\ I_{CEO} &= (1 + \beta) I_{CBO} \\ I_C \simeq \beta I_B, \quad I_B \simeq (1 - \alpha) I_E, \quad I_E \simeq I_C\end{aligned}$$

2. 放大作用与电压增益

$$\begin{aligned}\Delta i_C &= \alpha \Delta i_E \\ \Delta v_o &= -\Delta i_C R_L \\ A_v &= \frac{\Delta v_o}{\Delta v_i} \\ A_v &= \frac{0.98 \text{ V}}{20 \text{ mV}} \approx 49\end{aligned}$$

3. 固定偏置共射电路静态工作点

$$\begin{aligned}I_{BQ} &= \frac{V_{BB} - V_{BEQ}}{R_b} \\ I_{CQ} &= \beta I_{BQ} + I_{CEO} \simeq \beta I_{BQ} \\ V_{CEQ} &= V_{CC} - I_{CQ} R_C \\ V_{CE} &= V_{CC} - i_C R_C \\ v_{BE} &= V_{BB} + v_s - i_B R_b\end{aligned}$$

4. 截止、放大、饱和判据

$$I_{CS} \simeq \frac{V_{CC}}{R_C}$$

$$\beta I_B < I_{CS} \Rightarrow \text{放大区}$$

$$\beta I_B \geq I_{CS} \Rightarrow \text{饱和区}$$

$$I_B \simeq 0 \Rightarrow \text{截止区}$$

$$u_o = V_{CC} - I_C R_C$$

5. h 参数小信号模型

$$v_{be} = h_{ie} i_b + h_{re} v_{ce}$$

$$i_c = h_{fe} i_b + h_{oe} v_{ce}$$

$$h_{ie} = \left. \frac{\partial v_{BE}}{\partial i_B} \right|_{v_{CE}}, \quad h_{re} = \left. \frac{\partial v_{BE}}{\partial v_{CE}} \right|_{i_B}$$

$$h_{fe} = \left. \frac{\partial i_C}{\partial i_B} \right|_{v_{CE}}, \quad h_{oe} = \left. \frac{\partial i_C}{\partial v_{CE}} \right|_{i_B}$$

$$v_{be} \simeq h_{ie} i_b, \quad i_c \simeq \beta i_b$$

6. BJT 输入电阻 r_{be}

$$r_{be} = r_{bb'} + (1 + \beta) r_{b'e}$$

$$r_{b'e} = \frac{V_T}{I_{EQ}}$$

$$V_T \simeq 26 \text{ mV} \quad (T = 300 \text{ K})$$

$$r_{be} \approx 200 \Omega + (1 + \beta) \frac{26 \text{ mV}}{I_{EQ}(\text{mA})}$$

7. 基本共射放大电路小信号指标

$$v_i = i_b(R_b + r_{be})$$

$$i_c = \beta i_b$$

$$v_o = -i_c(R_C // R_L)$$

$$A_v = \frac{v_o}{v_i} = -\frac{\beta(R_C // R_L)}{R_b + r_{be}}$$

$$\text{若 } R_b \gg r_{be}, \quad A_v \approx -\frac{\beta(R_C // R_L)}{r_{be}}$$

$$R_i = R_b // r_{be} \approx r_{be}, \quad R_o \approx R_C$$

8. 考虑信号源内阻时的源电压增益

$$A_{vs} = \frac{v_o}{v_s} = \frac{v_o}{v_i} \cdot \frac{v_i}{v_s} = A_v \frac{R_i}{R_s + R_i}$$

9. 分压式射极偏置电路静态工作点

$$V_{BQ} \simeq \frac{R_{b2}}{R_{b1} + R_{b2}} V_{CC}$$

$$I_{EQ} \simeq \frac{V_{BQ} - V_{BEQ}}{R_E}$$

$$I_{CQ} \simeq I_{EQ}, \quad I_{BQ} = \frac{I_{CQ}}{\beta}$$

$$V_{CEQ} = V_{CC} - I_{CQ}R_C - I_{EQ}R_E \simeq V_{CC} - I_{CQ}(R_C + R_E)$$

10. 分压式射极偏置电路小信号指标

$$R_b = R_{b1} // R_{b2}$$

$$v_i = i_b [r_{be} + (1 + \beta)R_E]$$

$$v_o = -\beta i_b (R_C // R_L)$$

$$A_v = -\frac{\beta(R_C // R_L)}{r_{be} + (1 + \beta)R_E}$$

$$R_i = R_b // [r_{be} + (1 + \beta)R_E], \quad R_o \approx R_C$$

若 R_E 被旁路电容交流短路, 则 $A_v \approx -\frac{\beta(R_C // R_L)}{r_{be}}$

11. 阻容耦合双电源射极偏置电路

$$I_E = \frac{V_{EE} - V_{BE}}{R_E}, \quad I_C \simeq I_E, \quad I_B = \frac{I_C}{\beta}$$

$$V_{CE} = V_{CC} + V_{EE} - I_C(R_C + R_E)$$

$$A_v = -\frac{\beta(R_C // R_L)}{r_{be} + (1 + \beta)R_E}$$

$$R_i = R_b // [r_{be} + (1 + \beta)R_E], \quad R_o \approx R_C$$

12. 共集电极放大电路 (射极输出器)

$$V_{CC} = I_{BQ}R_b + V_{BEQ} + I_{EQ}R_E, \quad I_{EQ} = (1 + \beta)I_{BQ}$$

$$I_{BQ} = \frac{V_{CC} - V_{BEQ}}{R_b + (1 + \beta)R_E}, \quad I_{CQ} = \beta I_{BQ}$$

$$V_{CEQ} = V_{CC} - I_{EQ}R_E$$

$$R'_L = R_E // R_L$$

$$v_i = i_b [r_{be} + (1 + \beta)R'_L], \quad v_o = (1 + \beta)i_b R'_L$$

$$A_v = \frac{(1 + \beta)R'_L}{r_{be} + (1 + \beta)R'_L} = \frac{R'_L}{\frac{r_{be}}{1 + \beta} + R'_L} \approx 1$$

$$R_i = R_b // [r_{be} + (1 + \beta)R'_L]$$

$$R_o = R_E // \frac{R_s + r_{be}}{1 + \beta}$$

13. 共基极放大电路

$$A_v = \frac{v_o}{v_i} = \frac{\beta(R_C // R_L)}{r_{be}}$$
$$R_i = R_E // \frac{r_{be}}{1 + \beta}$$
$$R_o \approx R_C$$

共基电路电压增益为正，输入电阻小，输出电阻约为集电极电阻。

14. 三种基本组态的对比

组态	电压增益	输入电阻	输出电阻
共射	$-\frac{\beta(R_C // R_L)}{r_{be}}$	$R_b // r_{be}$	R_C
共集	$\frac{(1 + \beta)R'_L}{r_{be} + (1 + \beta)R'_L} \approx 1$	$R_b // [r_{be} + (1 + \beta)R'_L]$	$R_E // \frac{R_s + r_{be}}{1 + \beta}$
共基	$\frac{\beta(R_C // R_L)}{r_{be}}$	$R_E // \frac{r_{be}}{1 + \beta}$	R_C

15. 组合放大电路

$$A_v = A_{v1}A_{v2} \cdots A_{vn}$$
$$R_i = R_{i1}, \quad R_o = R_{on}$$
$$A_{v1} = -\frac{\beta_1 R'_{i2}}{r_{be1}}, \quad A_{v2} = -\frac{\beta_2(R_C // R_L)}{r_{be2} + R'_E}$$
$$A_v = A_{v1}A_{v2}$$

16. 复合管等效参数

$$\begin{aligned}r_{be} &= r_{be1} + (1 + \beta_1)r_{be2} \\ \beta &\simeq \beta_1\beta_2 \\ A_v &= \frac{(1 + \beta)R'_L}{r_{be} + (1 + \beta)R'_L} \approx 1 \\ R_i &= R_{b1} // R_{b2} // [r_{be} + (1 + \beta)R'_L] \\ R_o &= R_E // \frac{R_s + r_{be}}{1 + \beta}\end{aligned}$$

17. 频率响应

$$\begin{aligned}|A_v| &= |A_{vm}| \frac{1}{\sqrt{1 + \left(\frac{f_L}{f}\right)^2}} \frac{1}{\sqrt{1 + \left(\frac{f}{f_H}\right)^2}} \\ |A_v| &\approx \frac{|A_{vm}|}{\sqrt{1 + \left(\frac{f_L}{f}\right)^2}} \quad (f \ll f_H) \\ |A_v| &\approx \frac{|A_{vm}|}{\sqrt{1 + \left(\frac{f}{f_H}\right)^2}} \quad (f \gg f_L) \\ f = f_L \text{ 或 } f = f_H &\Rightarrow |A_v| = \frac{|A_{vm}|}{\sqrt{2}} \\ 20 \lg \frac{|A_v|}{|A_{vm}|} &= -3 \text{ dB}\end{aligned}$$